THANG HOANG, PhD

Contact Information

VT KnowledgeWorks II, Suite 2216
2202 Kraft Drive
Blacksburg, VA, 24060

Email: thanghoang@vt.edu
Cell: (+1) 541 – 230 – 8807
Webpage: http://thanghoang.github.io/

Research Interests

Applied Cryptography
Security and Privacy
Privacy-Enhancing Technologies
Biometrics

Education

University of South Florida, Tampa, Florida, United States — 2019 – 2020

Doctor of Philosophy, Computer Science
- Dissertation: *Privacy-Preserving and Functional Information Systems*
- Advisor: Dr. Attila Altay Yavuz

Oregon State University, Corvallis, Oregon, United States — 2015 – 2018

PhD student, Computer Science
- Advisor: Dr. Attila Altay Yavuz

Phi Kappa Phi Honor Society (OSU Chapter)

Chonnam National University, Gwangju, South Korea — 2012 – 2014

Master of Science, Computer Science
- Thesis: *Gait Authentication on Mobile Phone Using Pattern Recognition and Biometric Cryptosystem*
- Advisor: Dr. Deokjai Choi

University of Science VNU-HCMC, Ho Chi Minh city, Vietnam — 2006 – 2010

Bachelor of Science, Computer Science
- Thesis: *Integrating Camera Based Supervision System to Access Control Devices and Applications*
- Advisor: Dr. Thuc D. Nguyen

Professional Experiences

Assistant Professor, Department of Computer Science, Virginia Tech, Blacksburg, VA — Jan 2021 – current

Postdoctoral Fellow, Computer Science, Carnegie Mellon University, Pittsburgh, PA — Aug 2020 – Dec 2020
- Host: Prof. Elaine Shi

Research Associate, CSE, University of South Florida, Tampa, FL — Aug 2020 – Dec 2020
- Host: Prof. Attila A. Yavuz

Research Intern, Robert Bosch Research & Technology Center, Pittsburgh, PA, USA — 2016, 2018
- Mentor: Dr. Jorge Guajardo Merchan

Lecturer, Saigon Technology University, Ho Chi Minh city, Vietnam — 2014 – 2015

Research Scientist, RedOne Technologies Co., Ltd., Gwangju, South Korea — 2014 – 2015

Grants

New Cryptographic Audit Tools for Effective Data Integrity Attestation in Large-scale Storage-as-a-service Infrastructure — July 2022 – Jun 2023

PI(s): Thang Hoang (sole)
Sponsor: Commonwealth Cyber Initiative Southwest Virginia (CCI SWVA), Seeding Funding Program
Total: $20,000

Verifiable and Privacy-Preserving Machine Learning as a Service Jun 2022 – Aug 2022

PI(s): Thang Hoang (sole)
Sponsor: Commonwealth Cyber Initiative Southwest Virginia (CCI SWVA), Research Engagement Program (REP)
Total: $15,000

Towards Privacy-Enhancing Technologies Dec 2020 – current

PI(s): Thang Hoang (sole)
Sponsor: Robert Bosch, unrestricted gift
Total: $100,000

ACSAC Student Conferenceship Travel Grant 2016

Total: $1,000

Publications

**Underlined indicates my current/former advisees at the time of initial/completion.*

Journals

[1] Thao M. Dang, Thuc D. Nguyen, **Thang Hoang**, Hyunseok Kim, Andrew Beng Jin Teoh, and Deokjai Choi, "AVET: A Novel Transform Function To Improve Cancellable Biometrics Security". *IEEE Transactions on Information Forensics and Security* (IEEE TIFS), 2022.

[2] Lam Tran, **Thang Hoang**, Thuc Nguyen, Hyunil Kim, and Deokjai Choi, "Multi-Model Long Short-Term Memory Network for Gait Recognition Using Window-Based Data Segment". *IEEE Access*, Volume 9, pp. 23826–23839, February 2021.
doi:10.1109/ACCESS.2021.3056880

[3] **Thang Hoang**, Attila A. Yavuz, and Jorge Guajardo, "A Multi-server ORAM Framework with Constant Client Bandwidth Blowup". *ACM Transactions on Privacy and Security* (ACM TOPS), Volume 23, Issue 1, pp. 1–35, February 2020.
doi:10.1145/3369108

[4] **Thang Hoang**, Attila A. Yavuz, and Jorge Guajardo, "A Secure Searchable Encryption Framework for Privacy-Critical Cloud Storage Services". *IEEE Transactions on Services Computing* (IEEE TSC), Volume 14, Issue 6, pp. 1675–1689, November 2021 (pre-print published 2019).
doi:10.1109/TSC.2019.2897096

[5] **Thang Hoang**, Ceyhun D. Ozkaptan, Gabriel Hackebeil, and Attila A. Yavuz, "Efficient Oblivious Data Structures for Database Services on the Cloud". *IEEE Transactions on Cloud Computing* (IEEE TCC), Volume 9, Issue 2, pp. 598–609, April 2021 (pre-print published 2019).
doi:10.1109/TCC.2018.2879104

[6] **Thang Hoang**, Attila A. Yavuz, F. Betül Durak, and Jorge Guajardo, "A Multi-Server Oblivious Dynamic Searchable Encryption Framework". *Journal of Computer Security* (JCS), IOS Press, Volume 27, Issue 6, pp. 649–676, 2019.
doi:10.3233/JCS-191300

[7] **Thang Hoang**, Thuc Nguyen, and Deokjai Choi, "Gait Authentication on Mobile Phone Using Biometric Cryptosystem and Fuzzy Commitment Scheme". *International Journal of Information Security* (IJIS), Volume 14, Issue 6, pp. 549–560, November 2015.
doi:10.1007/s10207-015-0273-1

[8] **Thang Hoang** and Deokjai Choi, "Secure and Privacy Enhanced Gait Authentication on Smart Phone". *The Scientific World Journal* (TSWJ), Volume 2014, May 2014.
doi:10.1155/2014/438254

[9] **Thang Hoang**, Thuc D. Nguyen, Chuyen Luong, Son Do, and Deokjai Choi, "Adaptive Cross-Device Gait Recognition Using Mobile Accelerometer". *Journal of Information Processing System* (JIPS), Volume 9, Issue 2, pp. 333–348, June 2013.
doi:10.3745/JIPS.2013.9.2.333

[10] Viet Q. Vo, **Thang Hoang**, and Deokjai Choi, "Personalization in Mobile Activity Recognition System using K-Medoids Clustering Algorithm". *International Journal of Distributed Sensor Networks* (IJDSN), Volume 2013,

June 2013.
doi:10.1155/2013/315841

Conferences

[11] Haodi Wang and **Thang Hoang**, "ezDPS: An Efficient and Zero-Knowledge Machine Learning Inference Pipeline", in Privacy Enhancing Technologies Symposium (PETS), July 2023, Lausanne, Switzerland.

[12] Tung Le, Pengzhi Huang, Attila A. Yavuz, Elaine Shi, and **Thang Hoang**, "Efficient Dynamic Proof of Retrievability for Cold Storage", in the Annual Network and Distributed System Security Symposium (NDSS), February 2023, San Diego, CA, USA.

[13] Mohit Bhasi Thazhath, Jan Michalak, and **Thang Hoang**, "Harpocrates: Privacy-Preserving and Immutable Audit Log for Sensitive Data Operations", in *4th* IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS), December 2022 (virtual).

[14] Hai-Van Dang, Tran Phuong, Thuc Nguyen, and **Thang Hoang**, "ZAC: Efficient Zero-Knowledge Dynamic Universal Accumulator and Application to Zero-Knowledge Elementary Database", in *4th* IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS), December 2022 (virtual).

[15] Attila A Yavuz, Saif Nouma, **Thang Hoang**, Duncan Earl, and Scott Packard, "Distributed Cyber-infrastructures and Artificial Intelligence in Hybrid Post-Quantum Era", in *4th* IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS), December 2022 (virtual).

[16] Jiaheng Zhang, Tiancheng Xie, **Thang Hoang**, Elaine Shi, and Yupeng Zhang, "Polynomial Commitment with a One-to-Many Prover and Applications", in *31st USENIX Security Symposium* (USENIX Security), August 2022, Boston, MA, USA.

[17] Weikeng Chen, **Thang Hoang**, Jorge Guajardo, and Attila A. Yavuz, "Titanium: A Metadata-Hiding File-Sharing System with Malicious Security", in *the Annual Network and Distributed System Security Symposium* (NDSS), February 2022, San Diego, CA, USA.
doi:10.14722/ndss.2022.24161

[18] Efe Ulas Akay Seyitoglu, Attila A. Yavuz, and **Thang Hoang**, "Proof-of-Useful-Randomness: Mitigating the Energy Waste in Blockchain Proof-of-Work", in *18th International Conference on Security and Cryptography* (SECRYPT), July 2021 (virtual).
doi:10.5220/0010519204120419

[19] **Thang Hoang**, Rouzbeh Behnia, Yeongjin Jang, and Attila A. Yavuz, "MOSE: Practical Multi-User Oblivious Storage via Secure Enclaves", in *10th ACM Conference on Data and Application Security and Privacy* (CODASPY), March 2020, New Orleans, LA.
doi:10.1145/3374664.3375749

[20] **Thang Hoang**, Jorge Guajardo, and Attila A. Yavuz, "MACAO: A Maliciously-Secure and Client-Efficient Active ORAM Framework", in *the Annual Network and Distributed System Security Symposium* (NDSS), February 2020, San Diego, CA, USA.
doi:10.14722/ndss.2020.24313

[21] **Thang Hoang**, Muslum O. Ozmen, Yeongjin Jang, and Attila A. Yavuz, "Hardware-Supported ORAM in Effect: Practical Oblivious Search and Update on Very Large Dataset", in *19th Privacy Enhancing Technologies Symposium* (PETS), July 2019, Stockholm, Sweden.
doi:10.2478/popets-2019-0010

[22] **Thang Hoang**, Attila A. Yavuz, F. Betül Durak, and Jorge Guajardo, "Oblivious Dynamic Searchable Encryption on Distributed Cloud Systems", in *32nd Annual IFIP WG 11.3 Conference on Data and Applications Security and Privacy* (IFIP DBSec), July 2018, Bergamo, Italy. ***Best Paper Award***
doi:10.1007/978-3-319-95729-6_8

[23] Muslum O. Ozmen, **Thang Hoang**, and Attila A. Yavuz, "Forward-Private Dynamic Searchable Symmetric Encryption with Efficient Search", in *IEEE International Conference on Communications* (IEEE ICC), May 2018, Kansas City, MO, USA.
doi:10.1109/ICC.2018.8422480

[24] Lam Tran, **Thang Hoang**, Thuc Nguyen, and Deokjai Choi, "Improving Gait Cryptosystem Security Using Gray Code Quantization and Linear Discriminant Analysis", in *International Conference on Information Security* (ISC), pp. 214–229, November 2017, Ho Chi Minh, Vietnam.
doi:10.1007/978-3-319-69659-1_12

[25] **Thang Hoang**, Ceyhun D. Ozkaptan, Attila A. Yavuz, Jorge Guajardo, and Tam Nguyen, "S^3ORAM: A Computation-Efficient and Constant Client Bandwidth Blowup ORAM with Shamir Secret Sharing", in *24th ACM Conference on Computer and Communications Security* (ACM CCS), pp. 491–505, October 2017, Dallas, TX, USA.
doi:10.1145/3133956.3134090

[26] **Thang Hoang**, Attila A. Yavuz, and Jorge Guajardo, "Practical and Secure Dynamic Searchable Encryption via Oblivious Access on Distributed Data Structure", in *32nd Annual Computer Security Applications Conference* (ACSAC), pp. 302–313, December 2016, Los Angeles, CA, USA.
doi:10.1145/2991079.2991088

[27] **Thang Hoang**, Deokjai Choi, and Thuc Nguyen, "On the Instability of Sensor Orientation in Gait Verification on Mobile Phone", in *12th International Conference on Security and Cryptography* (SECRYPT), pp. 148–159, July 2015, Colmar, France.
doi:10.5220/0005572001480159

[28] **Thang Hoang** and Deokjai Choi, "A Biometric Cryptosystem Using Gait Captured from Mobile Accelerometer", in *FTRA International Symposium on Ubiquitous Computing and Embedded Systems*, December 2013, Danang, Vietnam. ***Best Paper Award*** (extended version published in [8])

[29] **Thang Hoang**, Deokjai Choi, Viet Vo, Anh Nguyen, and Thuc Nguyen, "A Lightweight Gait Authentication on Mobile Phone Regardless of Installation Error", in *28th International Information Security and Privacy Conference* (IFIP SEC), pp. 83–101, July 2013, Auckland, New Zealand.
doi:10.1007/978-3-642-39218-4_7

[30] Chuyen Luong, Son Do, **Thang Hoang**, and Deokjai Choi, "A Mobility Prediction Algorithm for The Seamless Handoff", in *5th International Conference Ubiquitous and Future Networks* (IEEE ICUFN), pp. 424–429, July 2013, Danang, Vietnam.
doi:10.1109/ICUFN.2013.6614854

[31] Viet Q. Vo, **Thang Hoang**, and Deokjai Choi, "Adaptive Energy-Saving Strategy for Activity Recognition on Mobile Phone", in *IEEE International Symposium on Signal Processing and Information Technology* (IEEE ISSPIT), pp. 95–100, December 2012, Ho Chi Minh city, Vietnam.
doi:10.1109/ISSPIT.2012.6621267

[32] **Thang Hoang**, Viet Q. Vo, Thuc D. Nguyen, and Deokjai Choi, "Gait Identification Using Accelerometer on Mobile Phone", in *1st International Conference on Control, Automation and Information Sciences* (IEEE ICCAIS), pp. 344–348, November 2012, Ho Chi Minh, Vietnam. ***Best Paper Award***
doi:10.1109/ICCAIS.2012.6466615

Patents

[1] **Thang Hoang** and Jorge Guajardo, "Secure and Efficient Multi-server Oblivious Random Access Machine in a Malicious Execution Environment", US Patent US20210135850A1, Filed: Nov 06, 2019 Issued: May 06, 2021.

[2] Attila A. Yavuz, Jorge Guajardo, and **Thang Hoang**, "Method and System for Search Pattern Oblivious Dynamic Symmetric Searchable Encryption", US Patent US11144663B2, Filed: Dec 28, 2017, Issued: Oct 21, 2021.

[3] **Thang Hoang**, Muslum O. Ozmen, and Attila A. Yavuz, "Forward-Private Dynamic Searchable Symmetric Encryption with Efficient Search", US Patent US10922273B1, Filed: Oct 10, 2017, Issued: Feb 16, 2021.

[4] Deokjai Choi, **Thang Hoang**, Thuc D. Nguyen, and Thu D. Tran, "Device and Method for Authentication System using Prime Number", Korea Patent 10-1754796, Filed: September 17, 2014, Issued: June 30, 2017.

[5] Deokjai Choi and **Thang Hoang**, "Secure Authentication System, and its Device and Method for Biometric Information, Derived Information from User Characteristic Information", Korea Patent 10-1622253, Filed: July 09, 2014, Issued: May 12, 2016.

[6] **Thang Hoang**, Deokjai Choi, and Chilwoo Lee, "Gait Authentication System and its Device and Method", Korea Patent 10-1622252, Filed: December 11, 2013, Issued: May 12, 2016.

Awards and Honors

- Best Paper Award in IFIP DBSec 2018, Bergamo, Italy 2018
- Best Paper Award in UCES 2013 symposium, Da Nang, Vietnam 2013
- Best Paper Award for the most innovative application in ICCAIS 2012, Ho Chi Minh city, Vietnam 2012

Professional Services

Organizing Committee

- IEEE S&P (Short Talks Chair) (2022)

Program Committee

- PETS (2023), WWW (2022), ACSAC (2021-2022), IEEE ICCCN (2021), CosDEO (PerCom Workshop) (2018, 2020)

Journal Reviewer

- IEEE Transactions on Services Computing (2022), ACM Digital Threats: Research and Practice (2020, 2021), ACM Transactions on Privacy and Security (2017, 2020, 2022), IEEE Transactions on Dependable and Secure Computing (2017 - 2022), IEEE Transactions on Information Forensics and Security (2018 - 2022), IEEE Transactions on Cybernetics (2019), IEEE Transactions on Cloud Computing (2019, 2021), Elsevier Computers & Security (2020), Elsevier Information Sciences (2017), Elsevier Journal of Information Security and Applications (2019)

Conference Reviewer

- ACM CCS (2021), IEEE S&P (2021), PETS (2020 - 2022), ACSAC (2017 - 2019), ASIACRYPT (2018), EUROCRYPT (2018), IEEE CSF (2021), IEEE CNS (2019), IFIP DBSec (2018, 2019), WiSec (2020), WWW (2019)

Talks

Privacy-Preserving Collaborative Information Processing

- Research Seminar at Robert Bosch RTC Remote, Feb 2022

Privacy-Preserving and Functional Information Systems

- Research Seminar at Robert Bosch RTC Remote, Jan 2021
- Seminar at Binghamton University Binghamton, NY, Mar 2020
- Seminar at Virginia Tech Remote, Mar 2020
- Seminar at University of Iowa Remote, Apr 2020
- Seminar at University of South Florida Tampa, FL, Apr 2020

MACAO: A Maliciously-Secure and Client-Efficient Active ORAM Framework

- Presented at ISOC NDSS 2020 San Diego, CA, USA, Feb 2020

Distributed ORAM for Data Outsourcing

- Seminar at Cornell University Ithaca, NY, USA, Nov 2019

S^3ORAM: A Computation-Efficient and Constant Client Bandwidth Blowup ORAM with Shamir Secret Sharing

- Presented at ACM CCS 2017 Dallas, TX, USA, Oct 2017

Practical and Secure Dynamic Searchable Encryption via Oblivious Access on Distributed Data Structure

- Presented at ACSAC 2016 Los Angeles, CA, USA, Dec 2016

Wireless Network Security

- Seminar at Eduroam Workshop, University of Indonesia Depok, Indonesia, Aug 2015

On the Instability of Sensor Orientation in Gait Verification on Mobile Phone

- Presented at SECRYPT 2015 Colmar, France, Jul 2015

A Lightweight Gait Authentication on Mobile Phone Regardless of Installation Error

- Presented at IFIP SEC 2013 Auckland, New Zealand, Jul 2013

Gait Identification Using Accelerometer on Mobile Phone

- Presented at IEEE ICCAIS 2012 Ho Chi Minh city, Vietnam, Nov 2012